This ICCV Workshop paper is the Open Access version, provided by the Computer Vision Foundation.
Except for this watermark, it is identical to the accepted version;
the final published version of the proceedings is available on IEEE Xplore.

FreqPure: a High-frequency Preservation Diffusion-based Purification Method for Protective Perturbation Removal

Yan Ju[1], Hongfei Xue[2], Siwei Lyu[1]*
[1] University at Buffalo, State University of New York {yanju, siweilyu}@buffalo.edu
[2] University at North Carolina at Charlotte hongfei.xue@charlotte.edu

Abstract

Diffusion models can be used to generate highly realistic likenesses of individuals when fine-tuned on a few personal images, raising serious concerns about unauthorized use and identity exploitation. In response, proactive defenses have been proposed by incorporating imperceptible protective perturbations into images to prevent unauthorized fine-tuning. Although these techniques are promising, their robustness under real-world conditions, such as common post-processing and adversarial purification, remains largely unexplored. In this work, we first evaluate the robustness of state-of-the-art protective methods with various post-processing and purification techniques. Moreover, to address the limitations of existing purification methods of removing protective perturbations along with natural high-frequency components, we propose FreqPure, a high-frequency-aware, diffusion-based purification approach that incorporates frequency consistency constraint to better preserve image fidelity during purification. It integrates a two-stage purification pipeline: 1) A reconstruction module that removes artifacts introduced by protection techniques. 2) A diffusion-based model that synthesizes high-frequency components, ensuring the output remains realistic. Extensive experiment results show that the proposed approach can purify perturbation on the image efficiently while preserving the natural high-frequency details of the images.

1. Introduction

In recent years, diffusion models have achieved remarkable success in tasks such as image synthesis, editing, and multi-modal generation [12, 22, 32]. Among them, Stable Diffusion [27], a multi-modal generative model built on Latent Diffusion Model (LDM) framework [27], has been widely used for artistic content generation [5, 42], visual storytelling [31, 46], personalized avatar generation [13, 19, 24, 38], etc. However, the increasing accessibility and powerful content control capabilities of Stable Diffusion [27] have raised serious concerns about privacy and copyright infringement. These concerns have been amplified by recent advances in fine-tuning techniques, such as DreamBooth [28] and LoRA (Low-Rank Adaptation) [14], which enable users to efficiently adapt pre-trained diffusion models using only a few images and limited computational resources. However, this capability introduces severe risks to personal privacy, including identity impersonation, deepfake generation, and misuse of sensitive imagery. These emerging risks have drawn increasing attention from both the public and the research community, as highlighted in recent media coverage and policy discussions [1, 6, 37].

In response, recent works have proposed protective perturbation techniques designed to safeguard image content from unauthorized use in diffusion-based models [8, 10, 20, 21, 29, 30, 34, 39–41, 45]. These approaches aim to introduce carefully crafted perturbations into images such that the protected images remain visually similar to the originals but degrade the performance of downstream diffusion models, thereby preventing misuse. While these methods show promising performance in controlled settings, an important and practical concern remains largely underexplored: the robustness of protective perturbations under real-world transformations, such as compression, blurring, or purification techniques that attempt to remove the protection. Moreover, existing diffusion-based purification methods are designed to remove protective perturbations via diffusion process [26, 44]. While effective at eliminating adversarial perturbations, this process often results in oversmoothing, producing purified images that lack important high-frequency details essential for visual realism and semantic fidelity, such as textures, wrinkles, hair patterns, and fine edges. More critically, if these overly smoothed images are used to fine-tune downstream diffusion models, the absence of high-frequency components can be propagated, ultimately degrading the quality and diversity of the generated outputs.

To address these limitations, in this work, we first eval-

*Corresponding author

uates the robustness of protective perturbations through three key strategies: 1) Common post-processing operations (e.g., JPEG compression, Gaussian blurring), 2) Fine-tuning diffusion models using various adaptation techniques (such as DreamBooth [28] and LoRA [14]), and 3) State-of-the-art adversarial purification methods (DiffPure [26] and GridPure [44]), which attempt to remove perturbations before downstream use. Our findings show that current purification techniques often degrade image by removing both protective perturbations and important natural details, especially high-frequency components critical for texture and edges. To overcome these challenges, we propose a novel frequency-aware purification framework, named FreqPure, which explicitly targets the removal of protective perturbations while preserving the original high-frequency content of the image. FreqPure introduces a two-stage purification pipeline: 1) A reconstruction module that keeps image content and reduces artifacts introduced by perturbation-based protection methods, serving as a clean structural foundation; 2) A diffusion-based reconstruction model trained to synthesize and recover high-frequency details conditioned on the reconstructed low-frequency image. This model ensures that the final output maintains both visual realism and semantic fidelity to the original image. Our contributions can be summarized as follows:

- We evaluate the robustness of existing protection methods through three practical strategies designed to simulate real-world application scenarios.
- We propose FreqPure, a novel high-frequency-aware, diffusion-based purification method that explicitly targets the removal of protective perturbations while preserving the natural high-frequency details crucial for image realism and semantic fidelity.
- Extensive experimental results demonstrate that FreqPure effectively eliminates protective perturbations while retaining high-frequency components typically present in real images.

2. Related Works

Protective Perturbation of Image. Different fine-tuning techniques [9, 14, 18, 28] of Stable Diffusion [27] raise concerns around misuse of personal images. To address this, proactive protection methods are proposed to inject imperceptible adversarial perturbations into images to disrupt the fine-tuning process. AdvDM [45] is an end-to-end algorithm to generate the adversarial examples for diffusion models to protect image copyright. Anti-DreamBooth [34] incorporates the DreamBooth fine-tuning process of Stable Diffusion into its consideration. Other works [39, 41, 43, 45] have also explored similar adversarial protection strategies.

Diffusion-based Image Purification. Adversarial purification techniques have been shown to be effective in removing adversarial perturbations from adversarial examples in image classification tasks [3, 15, 26, 35]. GDMP [35] utilize guided diffusion model for purification to help protect classifiers from adversarial attacks. DiffPure [26] leverages unconditional diffusion models to purify adversarial inputs. More recently, GrIDPure [44] extends DiffPure [26] by introducing a grid-based strategy combined with small-step iterative purification. Although it evaluate the robustness of different protection methods, it lacks natural frequency components in the purified images, making them seem overly smooth. These purification techniques, particularly GrIDPure [44], pose significant challenges for protection methods. Therefore, evaluating the resilience of protection strategies under such purification attacks is crucial for assessing their real-world robustness.

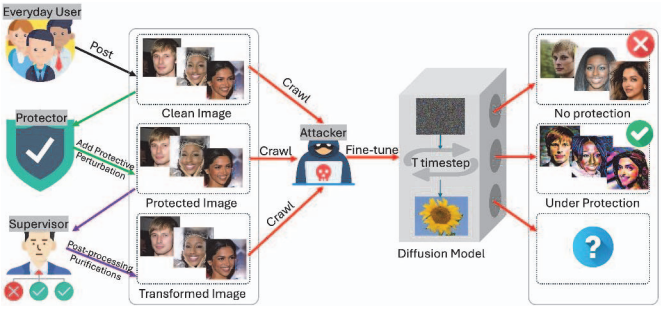


Figure 1. **Overview of the Protection Framework and Robustness Evaluation.**

3. Robustness Evaluation of Protection Methods

3.1. Problem Statement

Figure 1 show the scenario where the protection and the robustness testing are used. For everyday user, they upload personal images, such as portraits, to social media. Attackers can crawl these public images and create fake or harmful content by fine-tuning Stable Diffusion [27]. Protector embed imperceptible noise into images before posting to distort outputs when Attackers attempt fine-tuning. We act as a role of supervisor, testing whether protection remains effective under diverse transformations.

Attacker. In this process, the attacker aims to exploit publicly available images to generate personalized and highly similar visual content by fine-tuning the Stable Diffusion [27] model. The ultimate goal is to replicate or manipulate specific identities with high fidelity. To carry out this process, the attacker first gathers a small set of high-quality target images from the internet or social media platforms. Then, the attacker selects a fine-tuning technique, such as LoRA [14] or DreamBooth [28]. These methods enable the attacker to use diffusion model based on the text condition to control outputs with the original identity.

Input	+ Protection Method	+ Post-processing	PSNR (dB) ↑	SSIM ↑
Clean Image	N/A	N/A	/	/
	AdvDM [20]	N/A	38.48	0.94
		JPEG	35.94	0.92
		Blur	36.00	0.91
	AntiDB [34]	N/A	34.12	0.87
		JPEG	33.49	0.86
		Blur	34.09	0.87

Table 1. **Preservation of Visual Quality Under Different Protection and Post-Processing Strategies.**

Protector. To safeguard images from attacker, proactive defense embeds carefully designed imperceptible protective perturbations directly into the images to prevent these protected images from being effectively used to fine-tune generative models. In other words, if an attacker attempts to download and use such publicly available, perturbed images to train or personalize a diffusion model like Stable Diffusion [27], the resulting generated content will be significantly distorted.

Supervisor. Many of protection works offer limited robustness evaluations. To address this gap, in this work we act as a supervisor to assess the robustness of existing protective perturbation methods. The core objective of Supervisor is to simulate a wide range of real-world transformations, including commonly used post-processing such as Gaussian blur, JPEG compression, etc, which may be applied to images after they are uploaded to online platforms. Furthermore, several of purification approaches have shown potential in bypassing protective defenses, effectively restoring the usability of perturbed images for malicious fine-tuning. Thus, Supervisor considers purification methods, which are specifically designed to remove protective perturbations from images.

3.2. Protection Model

In our experiment, we select two state-of-the-art protective perturbation methods for evaluation: 1) AdvDM [20]: an end-to-end algorithm to generate the adversarial examples for diffusion models to protect image copyright. 2) Anti-DreamBooth (AntiDB) [34]: it incorporates the DreamBooth [28] fine-tuning process of Stable Diffusion [27] into its consideration. Both methods have demonstrated preliminary robustness in controlled settings, making them ideal candidates for more rigorous, real-world robustness testing.

3.3. Robustness Testing Strategy

In this work, we applies following three strategies to assess the robustness of protective methods under more challenging and representative conditions.

Different Fine-tuning method. Since the Protector has no prior knowledge of the specific fine-tuning strategy an attacker might use, it is critical that the added perturbations are robust to different fine-tuning techniques. In our work, we use two distinct fine-tuning strategies for Stable Diffusion [27]: 1) DreamBooth [28]: It is a widely used personalized fine-tuning method that injects instance-specific information into the model. It is particularly popular due to its exceptional quality and has become the core technique in many applications. 2) LoRA [14]: It uses low-rank weight updates to improve memory efficiency and is commonly used by the community. It only train weight increments in the cross attention layer of the UNet, enabling a quick and lightweight fine-tuning of the Diffusion model.

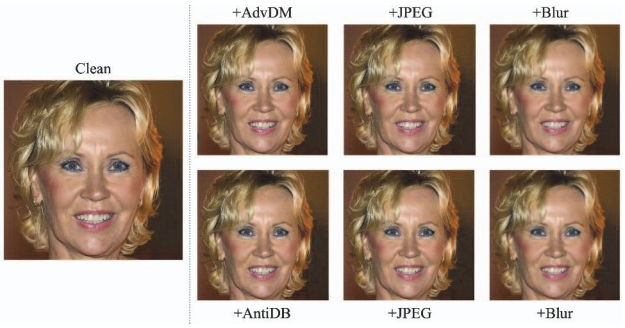


Figure 2. **Visual Comparison of Clean, Protected, and Post-processed Image. Zoom in for Best View.**

Real-world Post-processing. 1) JPEG Compression: Simulating common compression artifacts that occur when images are uploaded to social media. We set quality parameter as 70 in our experiments. 2) Gaussian Blurring: Reflecting common image editing operations, which can smooth out or weaken pixel-level perturbations. We set standard deviation parameter as 1.

Advanced Diffusion-based Purification. We used two diffusion-based purification methods to evaluate the robustness of existing protection methods. DiffPure [26] is an adversarial purification method that utilizes SDEdit [25] from an off-the-shelf unconditional diffusion model to purify adversarial images. GrIDPure [44] is a state-of-the-art purification method designed to preserve image resolution and intricate details.

3.4. Dataset and Metric

Dataset. Human facial images are among the most sensitive forms of personal data, as they can be used to identify individuals, infer demographic information, and even spoof biometric systems. Unauthorized use or leakage of such data poses serious privacy and security threats. Given this high sensitivity, we utilize a commonly used face dataset, CelebA-HQ [17] dataset, as the clean data following works [34, 44]. It is a high-quality version of CelebA [23] that consists of 30, 000 images at 1024×1024 resolution. We use a subset [4] and group images into 4 subjects with at least 15 images for each subject.

Metric. To evaluate the visual quality of the protected, post-processed, and purified images, we employ two widely used reference-based image quality metrics: Structural Similar-

Protection Method	LoRA [14]		DreamBooth [28]	
	FID ↑	IS ↓	FID ↑	IS ↓
Clean	158.10	0.92	121.26	0.91
Clean + AdvDM [20]	302.12	0.87	178.48	0.92
Clean + AntiDB [34]	294.67	0.87	145.08	0.92

Table 2. **Protection Effectiveness Across Different Fine-Tuning Methods.**

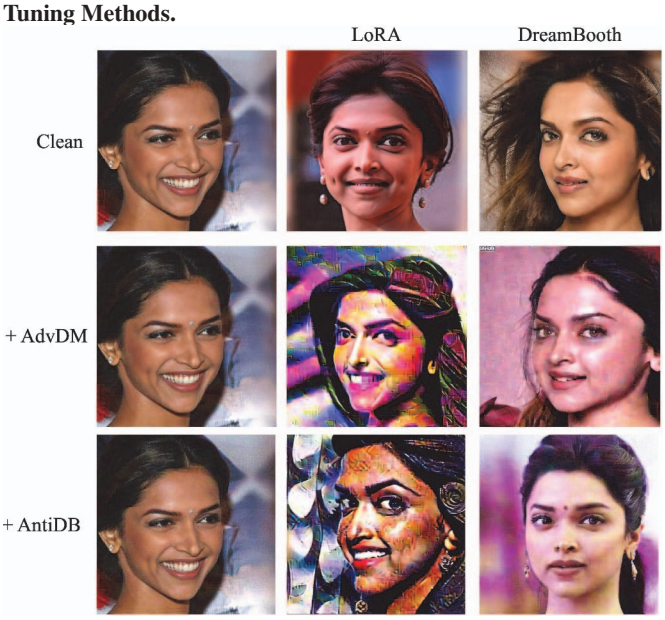


Figure 3. **Visual Quality of Images Generated by Stable Diffusion, Fine-tuned with Different Methods, using Clean and Protected Inputs.**

ity Index Measure (SSIM) and Peak Signal-to-Noise Ratio (PSNR). Both metrics are computed by comparing each processed image against its corresponding original (clean) image. We also assess the effectiveness of protection and purification methods by evaluating their impact on downstream diffusion model fine-tuning. Specifically, we use both the protected and purified images to fine-tune a diffusion model, and then evaluate the visual quality and the identity similarity of the generated images. We use the Fréchet Inception Distance (FID [11]) score, In our experiment, we use 5,000 face images from FFHQ dataset as the real images to calculate the FID score. We use another metric, Identity Similarity (IS), based on identity-specific metrics (ArcFace similarity [7]) for evaluation of personal identity protection. Specifically, we generate 50 personalized images for each subject using the fine-tuned diffusion model and use InceptionResnetV1 [33] model pretrained on vggface2 [2] to extract face embeddings and then compute the cosine similarity with the original clean image.

3.5. Visual Quality Evaluation

We compare the clean images with those protected using AdvDM [20] and AntiDB [34], both with and without subsequent post-processing operations (i.e., JPEG compression and Gaussian blur). This evaluation aims to assess the impact of the protection techniques and post-processing on the perceptual quality of the original images. Results are shown in Table 1. AdvDM [20] shows relatively high visual quality preservation, with a PSNR of 38.48 dB and SSIM of 0.94 in the unprocessed case, indicating that the protective perturbation is visually imperceptible. After JPEG compression and blur, PSNR slightly drops, suggesting that AdvDM [20] is robust to light post-processing while maintaining acceptable image quality. AntiDB [34] introduces more noticeable degradation, with an unprocessed PSNR of 34.12 dB and SSIM of 0.87, lower than AdvDM [20]. Post-processing with JPEG and blur results in further degradation, indicating that AntiDB [34]'s perturbation is more visually intrusive and more sensitive to downstream transformation.

+ Protection Method	+ Post-processing	DreamBooth [28]	
		FID ↑	IS ↓
AdvDM [20]	JPEG	138.15	0.89
	Blur	135.99	0.92
AntiDB [34]	JPEG	137.57	0.92
	Blur	122.58	0.92

Table 3. **Robustness of Protection Methods Against Post-processing.**

Figure 2 shows a qualitative comparison of visual quality across different conditions, including the original clean image, images protected by AdvDM [20] and AntiDB [34], and their respective versions after JPEG compression and Gaussian blur post-processing. This visual comparison is consistent with the results shown in Table 1. The clean image (top-left) serves as the reference, displaying natural textures, lighting, and sharpness. The image protected by AdvDM [20] (top row, middle-left) retains most of the original appearance with only slight perceptual deviations, demonstrating that the protection is applied without significantly degrading visual quality. However, when JPEG compression (top row, middle) is applied after AdvDM [20], visible compression artifacts such as blocking and loss of fine texture appear. Blurring (top row, right) further reduces detail, producing a smoother but less realistic image. In contrast, the image protected by AntiDB [34] (bottom row, middle-left) shows more noticeable artifacts even before post-processing, particularly around the facial features. When followed by JPEG compression (bottom row, middle), the image exhibits more pronounced distortions and reduced contrast. The blurred version (bottom row, right) similarly results in a loss of high-frequency details, leading to an unnatural and overly smooth appearance.

3.6. Robustness to Different Fine-tuning Method

Table 2 shows that both AdvDM [20] and AntiDB [34] introduce noticeable degradation in visual quality when used for fine-tuning Stable Diffusion [27] models with either

LoRA [14] or DreamBooth [28], which demonstrated their protection effectiveness. Under LoRA [14], both protection methods significantly increase FID scores, indicating a strong protection performance. For DreamBooth [28], both protection performance are lower than LoRA. Moreover, these protection methods demonstrate strong identity protection performance when fine-tuning is performed with LoRA [14], as evidenced by the lower identity similarity scores in the generated images compared to those from clean images. In contrast, when using DreamBooth [28] for fine-tuning, identity protection is less effective, as the generated images maintain high identity similarity, indicating that the subject's identity is still largely preserved.

Figure 3 presents a qualitative comparison of the visual quality of images generated by Stable Diffusion [27] models fine-tuned using LoRA [14] and DreamBooth [28], based on three types of input images: clean, clean + AdvDM [20] protection, and clean + AntiDB [34] protection. In the top row, we observe that both LoRA [14] and DreamBooth [28] produce high-quality and identity-preserving images when fine-tuned on clean inputs, confirming that in the absence of protection, both fine-tuning methods effectively learn the subject's appearance. In contrast, the middle and bottom rows illustrate the protection performance. When using AdvDM [20]-protected images as input (middle row), LoRA [14]-generated images exhibit severe visual distortions,, indicating a strong protection ability. However, under DreamBooth [28], the visual degradation is less extreme as the generated image still appears close to the original subject. This suggests DreamBooth [28] is more robust to AdvDM [20] protection, allowing partial recovery of identity. Similarly, the AntiDB [34]-protected images (bottom row) lead to even stronger distortions in LoRA [14] fine-tuning, where the generated outputs exhibit cartoonish textures and unnatural color patterns. These results indicate that AntiDB [34] provides strong identity protection in LoRA [14] by heavily disrupting the learning process. With DreamBooth [28], the generated images remain relatively high in visual quality and identity similarity, again reflecting DreamBooth [28]'s resilience to protection-based interference.

3.7. Robustness to Different Post-processing

Table 3 shows the evaluation results of protection robustness under post-processing conditions, where JPEG compression and Gaussian blur were applied to AdvDM [20]- and AntiDB [34]-protected images prior to DreamBooth [28] fine-tuning of the Stable Diffusion [27] model. This setup simulates common real-world image transformations and allows us to assess how well each protection method preserves its effectiveness under such conditions. From the table, we observe that AdvDM [20] shows strong protection under JPEG compression. Blur leads to a modest reduction in protection strength. For AntiDB [34], JPEG compression yields similar behavior. Notably, AntiDB [34] achieves the great robustness to blur post-processing. Overall, both AdvDM [20] and AntiDB [34] show robust performance under light JPEG and blur transformations.

+ Protection Method	+ Purification Method	SSIM ↑	PSNR (dB) ↑
AdvDM [20]	DiffPure [26]	0.33	28.09
	GrIDPure [44]	0.32	28.04
	FreqPure (Ours)	0.75	32.52
AntiDB [34]	DiffPure [26]	0.75	31.75
	GrIDPure [44]	0.82	28.76
	FreqPure (Ours)	0.86	32.96

Table 4. **Quantitative Analysis of Visual Quality Across Purification Methods.**

4. FreqPure

Existing diffusion-based purification methods primarily focus on reconstructing protected images using diffusion models. In this process, the diffusion model iteratively refines the image, gradually eliminating high-frequency components, including the added protective perturbations intended to protect sensitive information. While effective at removing such perturbations, these methods exhibit two significant limitations that hinder the overall image quality. First, the indiscriminate suppression of high-frequency components leads to the removal of not only the protective perturbations but also the natural, fine-grained details present in unaltered images. These high-frequency details, such as skin textures, hair strands, and facial wrinkles, are essential for preserving the authenticity and visual realism of the image. Due to the absence of natural high-frequency components, purified images often appear overly smooth, as shown in Figure 5. This phenomenon is also illustrated in Figure 6, where the DCT comparison highlights the loss of high frequency component in the purified images compared with clean image. Second, when these images are used to fine-tune downstream diffusion models, the resulting generations may lack the visual quality typically achieved with clean images. These limitations underscore the need for improved purification methods that can effectively remove protective perturbations while preserving the natural high-frequency details crucial for maintaining image fidelity and realism.

4.1. Framework

To address above issues in existing purification methods, our work proposes a high-frequency-aware, diffusion-based purification method. Figure 4 illustrates the training and inference workflow of our proposed purification framework, which aims to remove protective perturbations from images while preserving high-frequency details.

During training, we use original (clean) face images as

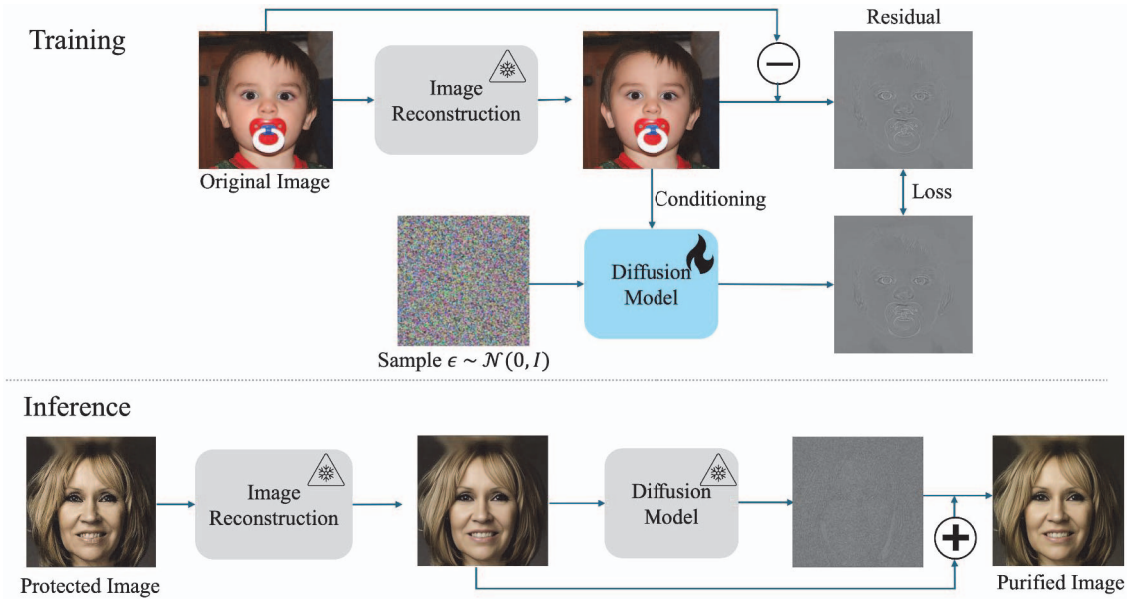


Figure 4. **Overview of the FreqPure Framework.**

input. The process begins by passing the original image through an Image Reconstruction module, which acts as a high-frequency protection removal and a visual artifacts removal, which preserve the base facial content. This output serves as a conditioning signal to guide the diffusion model. To train the model to learn the high-frequency residuals, we compute the difference between the original image and the low-frequency version produced by the restoration module. This residual represents the high-frequency details, which is used as the target output. The training then follows a typical DDPM [12] process. At inference time, a protected image (e.g., one protected by AdvDM [20] or AntiDB [34]) is first passed through the Image Reconstruction module, generating an clean image. This restored content is used as the conditioning input to the diffusion model, which then reconstructs the missing natural high-frequency residuals. These predicted residuals are subsequently added back to the restored base image to form the final purified output. The result is a high-fidelity image that preserves visual realism while eliminating protective perturbations.

4.1.1. Image Reconstruction Module

We utilize a publicly available face image restoration model GFPGAN [36] for several reasons: 1) Extract Semantic Content: GFPGAN [36] is designed to reconstruct the face images with high fidelity and natural appearance. By leveraging its learned priors on natural face structures, GFPGAN [36] can extract semantic content in the face images, and based on that, we can calculate the high-frequency residual for further step. 2) Artifact Reduction in Protected Images: In practice, images subjected to protection techniques (e.g., adversarial perturbations) often exhibit visible artifacts or subtle distortions, even when the protection

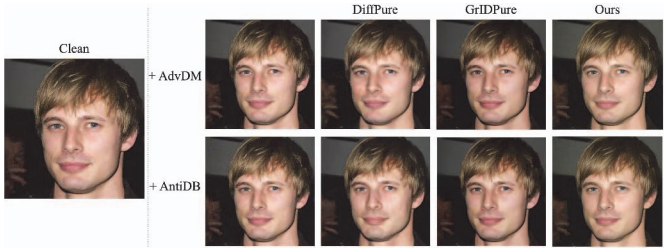


Figure 5. **Visual Comparison of Purification Methods. Zoom in for Best View.**

methods aim to constrain perturbations within a small magnitude. By applying GFPGAN [36], we can extract low-frequency image content and mitigate visible artifacts introduced by protection mechanisms. This restoration step not only improves the visual quality of the purified image but also provides residuals for the subsequent purification process.

Specifically, we first construct a training dataset consisting of 5,000 high-resolution face images (resolution: 1024 × 1024 pixels) sourced from FFHQ [16] dataset, a diverse set of publicly available face datasets. For each image in this dataset, we apply the GFPGAN model [36] to obtain its corresponding clean and without artifacts low-frequency image content. The difference between the original image and the GFPGAN [36]-processed image is computed pixel-wise, and we treat this residual as the ground-truth high-frequency component. This high-frequency residual captures the fine-grained details, such as edges, textures, etc.

4.1.2. High-Frequency Component Compensation

Following the first stage of our pipeline, we introduce the second stage: a conditioned diffusion model specif-

+ Protection Method	+ Purification Method	DreamBooth [28]	
		FID ↓	IS ↑
AdvDM [20]	DiffPure [26]	126.32	0.91
	GrIDPure [44]	138.9	0.93
	FreqPure (Ours)	110.62	0.94
AntiDB [34]	DiffPure [26]	124.44	0.92
	GrIDPure [44]	136.63	0.90
	FreqPure (Ours)	108.99	0.95

Table 5. **Effectiveness Comparison of Different Purification Methods.**

ically designed to recover and synthesize the natural high-frequency components of face images. The primary goal of this stage is to restore fine-grained details, that are typically lost in other purification methods or suppressed by the protective perturbations. Specifically, using the high-frequency residual as ground-truth, we design a conditional diffusion model that learns to reconstruct these components with low-frequency inputs as conditioning. The process follows the standard DDPM [12] process. During training, the model is tasked with predicting the high-frequency residuals given a randomly sampled noise and the corresponding low-frequency image as a conditioning signal. The model is trained to reconstruct the high-frequency residual through the loss function defined in Equation 1 to minimize the difference between the predicted and ground-truth residuals, where $\mathcal{L}_1$ loss is calculated using $\mathcal{L}_1(\theta, x_0) = \mathbb{E}_{x_0,t,c,\epsilon\sim\mathcal{N}(0,I)} \|\epsilon - \epsilon_\theta(x_{t+1}, t, c)\|_1$, mean square loss $\mathcal{L}_{MSE}$ is calculated using $\mathcal{L}_{MSE}(\theta, x_0) = \mathbb{E}_{x_0,t,c,\epsilon\sim\mathcal{N}(0,I)} \|\epsilon - \epsilon_\theta(x_{t+1}, t, c)\|_2^2$, and we also use a frequency loss to better learn the frequency distribution of the residual, denoted as $\mathcal{L}_{freq}$. It is calculated using $\mathcal{L}_{freq}(\theta, x_0) = \mathbb{E}_{x_0,t,c,\epsilon\sim\mathcal{N}(0,I)} \|fft(\epsilon) - fft(\epsilon_\theta(x_{t+1}, t, c))\|_1$.

$$\mathcal{L}(\theta, x_0) = \alpha\mathcal{L}_1(\theta, x_0) + \beta\mathcal{L}_{MSE}(\theta, x_0) + \gamma\mathcal{L}_{freq}(\theta, x_0) \tag{1}$$

Through this learning process, the diffusion model acquires the capability to synthesize plausible and diverse high-frequency details, such as skin textures, hair strands, and edge facial details. The model effectively bridges the gap introduced by traditional purification mechanisms, ensuring that the purified images retain both visual fidelity and natural frequency distributions.

5. Experiments

In our experiments, we adopt the same datasets and evaluation metrics as described in Section 3.4 to ensure a consistent and fair comparison across methods. We design our evaluation around two key aspects: 1) image quality preservation from spatial and frequency domain, and 2) purification effectiveness for downstream personalized diffusion fine-tuning.

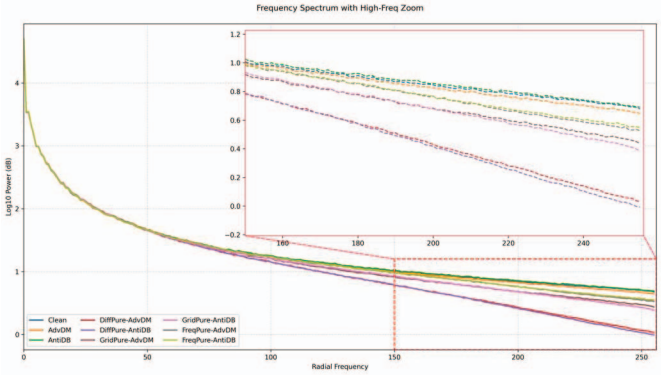


Figure 6. **Frequency Spectrum Analysis of Different Image Sets.**

5.1. Visual Quality Comparison

We compare the visual quality of the purified images against the corresponding clean (unaltered) images. Both spatial inspections and frequency-domain analyses (e.g., power spectrum comparisons using DCT) are performed to validate the high-frequency recovery capabilities of our model. Table 4 presents a quantitative evaluation of the visual quality preservation achieved by DiffPure [26], GrIDPure [44], and our proposed method. They are applied to images protected by AdvDM [20] and AntiDB [34], respectively. We report SSIM and PSNR as evaluation metrics, where higher values indicate better image quality and, consequently, more effective purification performance. For AdvDM-protected images, our method significantly outperforms both DiffPure and GrIDPure. This indicates that our method is substantially better at preserving structure and restoring high-quality details from the protected image. For AntiDB-protected images, our method again achieves the highest performance, clearly outperforming DiffPure and GrIDPure. These results demonstrate that our proposed purification method consistently delivers the best visual reconstruction quality for both protection methods.

Figure 5 shows a qualitative comparison of using different purification methods. For images protected by AdvDM [20] (top row), DiffPure [26] and GrIDPure [44] both succeed in partially removing artifacts introduced by the protection method. However, DiffPure [26] tends to retain some residual noise and lacks sharpness in facial details, while GrIDPure [44] results in an overly smoothed and slightly unnatural appearance, especially around the eyes and skin texture. In contrast, our method produces a visually cleaner and more realistic output, closely resembling the original clean image. Facial features such as the eyes, nose, and hairline are sharper and more natural, indicating more effective restoration of high-frequency details. For images protected by AntiDB [34] (bottom row), similar patterns are observed. These visual results are consistent with the quantitative evaluations presented in Table 4,

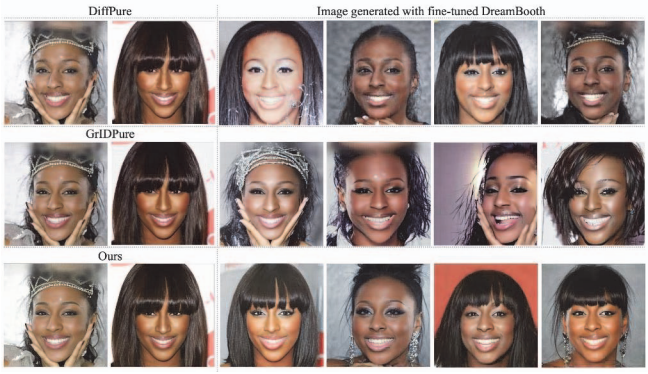


Figure 7. **Visual Comparison of Outputs from DreamBooth-Fine-Tuned Stable Diffusion Using Inputs from Various Purification Methods.**

further confirming that our purification method provides a better visual performance.

5.2. Frequency Domain Analysis

Figure 6 presents the average frequency spectrum analysis of different image sets using the Discrete Cosine Transform (DCT). The x-axis represents the frequency components, while the y-axis indicates power at each frequency level. A frequency curve closer to that of the clean images indicates better preservation. The AdvDM [20]- and AntiDB [34]-protected images deviate from the clean baseline slightly, especially in the mid-to-high frequency range, reflecting the expected suppression of high-frequency components due to perturbations. Among the purified outputs: The curves for DiffPure [26] and GrIDPure [44] show the largest deviation from the original frequency spectrum, particularly in the higher frequency regions. This indicates that while these methods remove the perturbations, they also excessively suppress natural high-frequency content, leading to over-smoothed images and loss of edges. In contrast, our method closely follows the frequency profile of the clean images, especially across mid and high-frequency ranges. This suggests that our method is more effective in recovering fine-grained image details, resulting in outputs that are perceptually sharper and more consistent to the original image distribution.

5.3. Effectiveness of Purification

We also evaluate whether purified images can serve as reliable inputs for fine-tuning diffusion models. Ideally, purified images should enable successful model fine-tuning, resulting in high-quality generations. Table 5 reports the effectiveness of different purification methods for DreamBooth [28] fine-tuning. It shows that our proposed purification method consistently outperforms both DiffPure and GrIDPure across all metrics. For both AdvDM and AntiDB protected images, our method achieves the lowest FID scores, indicating that the images generated from our purified images are closest in distribution to those generated from clean images. In terms of identity preservation, our method also yields the highest IS scores, suggesting that the generated images retain better identity similarity to the originals. These results demonstrate that our purification approach is the most effective in restoring protected images for downstream fine-tuning with DreamBooth, preserving image quality and identity similarity. The visual comparison in Figure 7 further demonstrates the superiority of our method.

6. Conclusion

In this work, we evaluate the robustness of existing image protection methods against common real-world post-processing techniques, including JPEG compression, Gaussian blur, and advanced purification methods. Our findings reveal that while these protection methods exhibit a certain degree of robustness, they remain vulnerable to specifically designed purification techniques. Moreover, we propose FreqPure, a diffusion-based purification framework to remove protective perturbation while preserving natual high-frequency components. We validate our approach through extensive quantitative and qualitative experiments. The results demonstrate that our method consistently outperforms existing baselines in terms of spatial and frequency analysis. Our proposed purification method can serve as a strong benchmark for evaluating the robustness of future protection techniques, and it can also be effectively used to remove adversarial perturbations and watermarks.

Limitation. Our work has two main limitations. First, we focus exclusively on face images, as they reveal identifiable features and are among the most susceptible to exploitation for generating fake or harmful content. Second, our method relies on the performance of the image content reconstruction model used in the first stage. With a strong reconstruction model, our approach can further enhance the high-frequency quality of the purified images.

Ethics Statement. While our method can effectively remove perturbations from images, its primary purpose is to serve as a strong benchmark for evaluating existing perturbation-based image protection techniques and to promote the development of improved protection methods. We acknowledge the potential risk that malicious actors could use our approach to remove protective perturbations and create more deceptive, personalized images. To mitigate such misuse, code and model weights will be shared only to trustworthy parties. We will release the code as open source under the condition that it must not be used to produce or distribute harmful, offensive, dehumanizing, or otherwise damaging representations of individuals, communities, cultures, religions, or their environments.

References

[1] BBC. "Art is dead Dude" - the rise of the AI artists stirs debate., 2022. https://www.bbc.com/news/technology-62788725. 1

[2] Qiong Cao, Li Shen, Weidi Xie, Omkar M Parkhi, and Andrew Zisserman. Vggface2: A dataset for recognising faces across pose and age. In *2018 13th IEEE international conference on automatic face & gesture recognition (FG 2018)*, pages 67–74. IEEE, 2018. 4

[3] Nicholas Carlini, Florian Tramer, Krishnamurthy Dj Dvijotham, Leslie Rice, Mingjie Sun, and J Zico Kolter. (certified!!) adversarial robustness for free! *arXiv preprint arXiv:2206.10550*, 2022. 2

[4] CelebA-HQ. CelebA-HQ, 2023. Available at https://github.com/ndb796/CelebA-HQ-Face-Identity-and-Attributes-Recognition-PyTorch. 3

[5] Zhiyuan Chen, Jiajiong Cao, Zhiquan Chen, Yuming Li, and Chenguang Ma. Echomimic: Lifelike audio-driven portrait animations through editable landmark conditions. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 2403–2410, 2025. 1

[6] CNN. AI won an art contest, and artists are furious., 2022. https://www.cnn.com/2022/09/03/tech/ai-art-fair-winner-controversy/index.html. 1

[7] Jiankang Deng, Jia Guo, Niannan Xue, and Stefanos Zafeiriou. Arcface: Additive angular margin loss for deep face recognition. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4690–4699, 2019. 4

[8] Jinhao Duan, Fei Kong, Shiqi Wang, Xiaoshuang Shi, and Kaidi Xu. Are diffusion models vulnerable to membership inference attacks? In *International Conference on Machine Learning*, pages 8717–8730. PMLR, 2023. 1

[9] Rinon Gal, Yuval Alaluf, Yuval Atzmon, Or Patashnik, Amit H Bermano, Gal Chechik, and Daniel Cohen-Or. An image is worth one word: Personalizing text-to-image generation using textual inversion. *arXiv preprint arXiv:2208.01618*, 2022. 2

[10] Rohit Gandikota, Joanna Materzynska, Jaden Fiotto-Kaufman, and David Bau. Erasing concepts from diffusion models. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2426–2436, 2023. 1

[11] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *Advances in neural information processing systems*, 30, 2017. 4

[12] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems*, 33:6840–6851, 2020. 1, 6, 7

[13] Susung Hong, Donghoon Ahn, and Seungryong Kim. Debiasing scores and prompts of 2d diffusion for robust text-to-3d generation. *arXiv preprint arXiv:2303.15413*, 2(5), 2023. 1

[14] Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, Weizhu Chen, et al. Lora: Low-rank adaptation of large language models. *ICLR*, 1(2):3, 2022. 1, 2, 3, 4, 5

[15] Wan Jiang, Yunfeng Diao, He Wang, Jianxin Sun, Meng Wang, and Richang Hong. Unlearnable examples give a false sense of security: Piercing through unexploitable data with learnable examples. In *Proceedings of the 31st ACM International Conference on Multimedia*, pages 8910–8921, 2023. 2

[16] Tero Karras, Samuli Laine, and Timo Aila. A style-based generator architecture for generative adversarial networks. In *CVPR*, pages 4401–4410, 2019. 6

[17] Tero Karras et al. Progressive growing of gans for improved quality, stability, and variation. *arXiv preprint arXiv:1710.10196*, 2017. 3

[18] Nupur Kumari, Bingliang Zhang, Richard Zhang, Eli Shechtman, and Jun-Yan Zhu. Multi-concept customization of text-to-image diffusion. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 1931–1941, 2023. 2

[19] Xiaoming Li, Xinyu Hou, and Chen Change Loy. When stylegan meets stable diffusion: a w+ adapter for personalized image generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 2187–2196, 2024. 1

[20] Chumeng Liang, Xiaoyu Wu, Yang Hua, Jiaru Zhang, Yiming Xue, Tao Song, Zhengui Xue, Ruhui Ma, and Haibing Guan. Adversarial example does good: Preventing painting imitation from diffusion models via adversarial examples. *arXiv preprint arXiv:2302.04578*, 2023. 1, 3, 4, 5, 6, 7, 8

[21] Hanwen Liu, Zhicheng Sun, and Yadong Mu. Countering personalized text-to-image generation with influence watermarks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 12257–12267, 2024. 1

[22] Yixin Liu, Kai Zhang, Yuan Li, Zhiling Yan, Chujie Gao, Ruoxi Chen, Zhengqing Yuan, Yue Huang, Hanchi Sun, Jianfeng Gao, et al. Sora: A review on background, technology, limitations, and opportunities of large vision models. *arXiv preprint arXiv:2402.17177*, 2024. 1

[23] Ziwei Liu, Ping Luo, Xiaogang Wang, and Xiaoou Tang. Deep learning face attributes in the wild. In *Proceedings of the IEEE international conference on computer vision*, pages 3730–3738, 2015. 3

[24] Yue Ma, Hongyu Liu, Hongfa Wang, Heng Pan, Yingqing He, Junkun Yuan, Ailing Zeng, Chengfei Cai, Heung-Yeung Shum, Wei Liu, et al. Follow-your-emoji: Fine-controllable and expressive freestyle portrait animation. In *SIGGRAPH Asia 2024 Conference Papers*, pages 1–12, 2024. 1

[25] Chenlin Meng, Yutong He, Yang Song, Jiaming Song, Jiajun Wu, Jun-Yan Zhu, and Stefano Ermon. Sdedit: Guided image synthesis and editing with stochastic differential equations. *arXiv preprint arXiv:2108.01073*, 2021. 3

[26] Weili Nie, Brandon Guo, Yujia Huang, Chaowei Xiao, Arash Vahdat, and Anima Anandkumar. Diffusion models for adversarial purification. *arXiv preprint arXiv:2205.07460*, 2022. 1, 2, 3, 5, 7, 8

[27] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image

synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 10684–10695, 2022. 1, 2, 3, 4, 5

[28] Nataniel Ruiz, Yuanzhen Li, Varun Jampani, Yael Pritch, Michael Rubinstein, and Kfir Aberman. Dreambooth: Fine tuning text-to-image diffusion models for subject-driven generation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 22500–22510, 2023. 1, 2, 3, 4, 5, 7, 8

[29] Hadi Salman, Alaa Khaddaj, Guillaume Leclerc, Andrew Ilyas, and Aleksander Madry. Raising the cost of malicious ai-powered image editing. *arXiv preprint arXiv:2302.06588*, 2023. 1

[30] Shawn Shan, Jenna Cryan, Emily Wenger, Haitao Zheng, Rana Hanocka, and Ben Y Zhao. Glaze: Protecting artists from style mimicry by {Text-to-Image} models. In *32nd USENIX Security Symposium (USENIX Security 23)*, pages 2187–2204, 2023. 1

[31] Xiaoqian Shen and Mohamed Elhoseiny. Storygpt-v: Large language models as consistent story visualizers. In *Proceedings of the Computer Vision and Pattern Recognition Conference*, pages 13273–13283, 2025. 1

[32] Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. *arXiv preprint arXiv:2011.13456*, 2020. 1

[33] Christian Szegedy, Sergey Ioffe, Vincent Vanhoucke, and Alexander Alemi. Inception-v4, inception-resnet and the impact of residual connections on learning. In *Proceedings of the AAAI conference on artificial intelligence*, 2017. 4

[34] Thanh Van Le, Hao Phung, Thuan Hoang Nguyen, Quan Dao, Ngoc N Tran, and Anh Tran. Anti-dreambooth: Protecting users from personalized text-to-image synthesis. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 2116–2127, 2023. 1, 2, 3, 4, 5, 6, 7, 8

[35] Jinyi Wang, Zhaoyang Lyu, Dahua Lin, Bo Dai, and Hongfei Fu. Guided diffusion model for adversarial purification. *arXiv preprint arXiv:2205.14969*, 2022. 2

[36] Xintao Wang, Yu Li, Honglun Zhang, and Ying Shan. Towards real-world blind face restoration with generative facial prior. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 9168–9178, 2021. 6

[37] WashingtonPost. He made a children's book using AI. Then came the rage., 2022. https://www.washingtonpost.com/technology/2023/01/19/ai-childrens-book-controversy-chatgpt-midjourney/. 1

[38] Huawei Wei, Zejun Yang, and Zhisheng Wang. Aniportrait: Audio-driven synthesis of photorealistic portrait animation. *arXiv preprint arXiv:2403.17694*, 2024. 1

[39] Ruijia Wu, Yuhang Wang, Huafeng Shi, Zhipeng Yu, Yichao Wu, and Ding Liang. Towards prompt-robust face privacy protection via adversarial decoupling augmentation framework. *arXiv preprint arXiv:2305.03980*, 2023. 1, 2

[40] Jingyao Xu, Yuetong Lu, Yandong Li, Siyang Lu, Dongdong Wang, and Xiang Wei. Perturbing attention gives you more bang for the buck: Subtle imaging perturbations that efficiently fool customized diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24534–24543, 2024.

[41] Xiaoyu Ye, Hao Huang, Jiaqi An, and Yongtao Wang. Duaw: Data-free universal adversarial watermark against stable diffusion customization. *arXiv preprint arXiv:2308.09889*, 2023. 1, 2

[42] Tianyi Zhang, Zheng Wang, Jing Huang, Mohiuddin Muhammad Tasnim, and Wei Shi. A survey of diffusion based image generation models: Issues and their solutions. *arXiv preprint arXiv:2308.13142*, 2023. 1

[43] Zhengyue Zhao, Jinhao Duan, Xing Hu, Kaidi Xu, Chenan Wang, Rui Zhang, Zidong Du, Qi Guo, and Yunji Chen. Unlearnable examples for diffusion models: Protect data from unauthorized exploitation. *arXiv preprint arXiv:2306.01902*, 2023. 2

[44] Zhengyue Zhao, Jinhao Duan, Kaidi Xu, Chenan Wang, Rui Zhang, Zidong Du, Qi Guo, and Xing Hu. Can protective perturbation safeguard personal data from being exploited by stable diffusion? In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24398–24407, 2024. 1, 2, 3, 5, 7, 8

[45] Boyang Zheng, Chumeng Liang, Xiaoyu Wu, and Yan Liu. Understanding and improving adversarial attacks on latent diffusion model. 2023. 1, 2

[46] Sixiao Zheng and Yanwei Fu. Contextualstory: Consistent visual storytelling with spatially-enhanced and storyline context. In *Proceedings of the AAAI Conference on Artificial Intelligence*, pages 10617–10625, 2025. 1